Thalassemia

(the genetic basis)

-Hemoglobinopathies: Mutations which cause a change in primary structure of one of the globin chains--over 700 known

-Rate of globin chain synthesis are theoretically normal.

-Thalassemias: Mutations which alter the level of expression of one of the globin chains--over 280 known

- Heritable, hypochromic anemias-varying degrees of severity
- Genetic defects result in decreased or absent globin chain synthesis
- High incidence in Asia, Africa, Mideast, and Mediterranean countries.

Hemoglobin

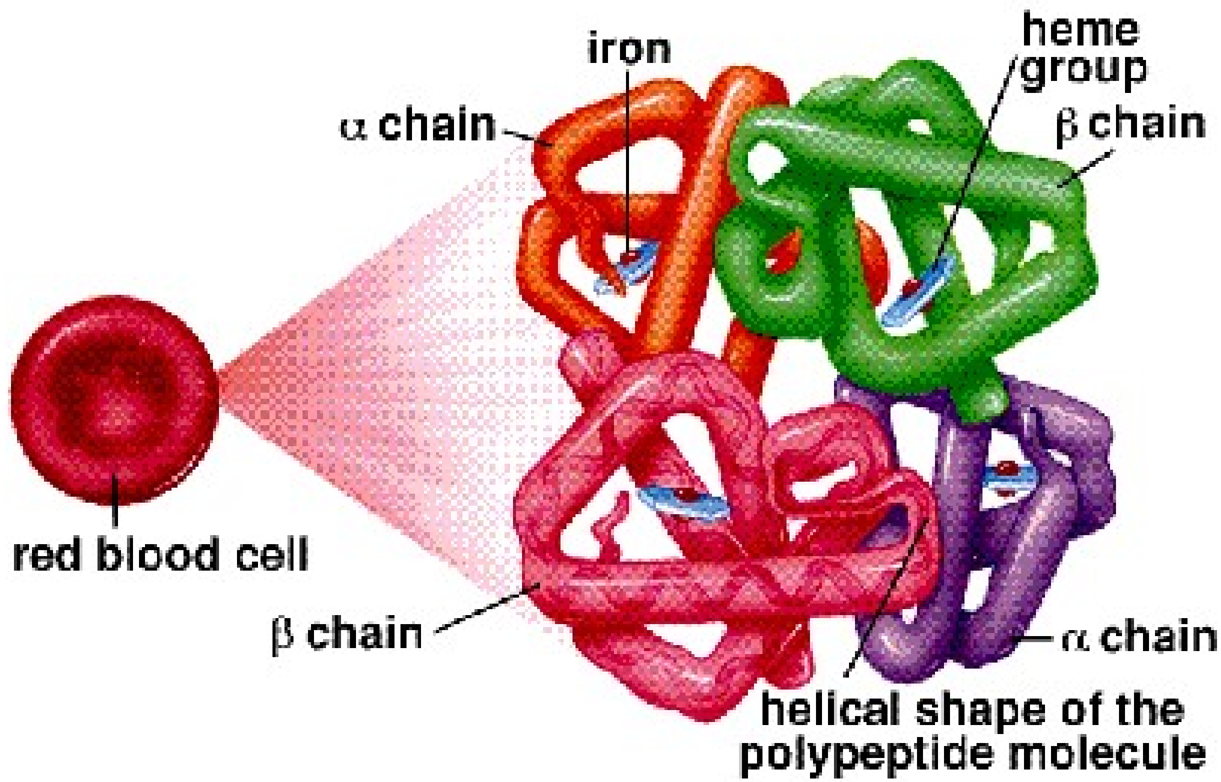
- Each complex consists of :
  - Four polypeptide chains, non-covalently bound

    four heme complexes with iron bound(Each globin chain bounds one heme group)

  - Four O2 binding sites

Hemoglobin Structure

- Four subunits
  - two α
  - two β
- Iron
- Heme
- Binds 4 O_2



Globin chains

- Alpha Globin
  - 141 amino acids
  - Coded for on Chromosome 16
  - Found in normal adult hemoglobin (A1, A2) F Hgb and Gower2
- Beta Globin
  - 146 amino acids
  - Coded for on Chromosome 11, found in Hgb A1 and Portland 2
- Delta Globin
  - Found in Hemoglobin A2--small amounts in all adults
- Gamma Globin
  - Found in Fetal Hemoglobin , and Portland 1
- Zeta Globin
  - Found in embryonic hemoglobin (Gower1 and Portland 1and 2)
- Epsilon
  - Found in Gower 1and Gower2

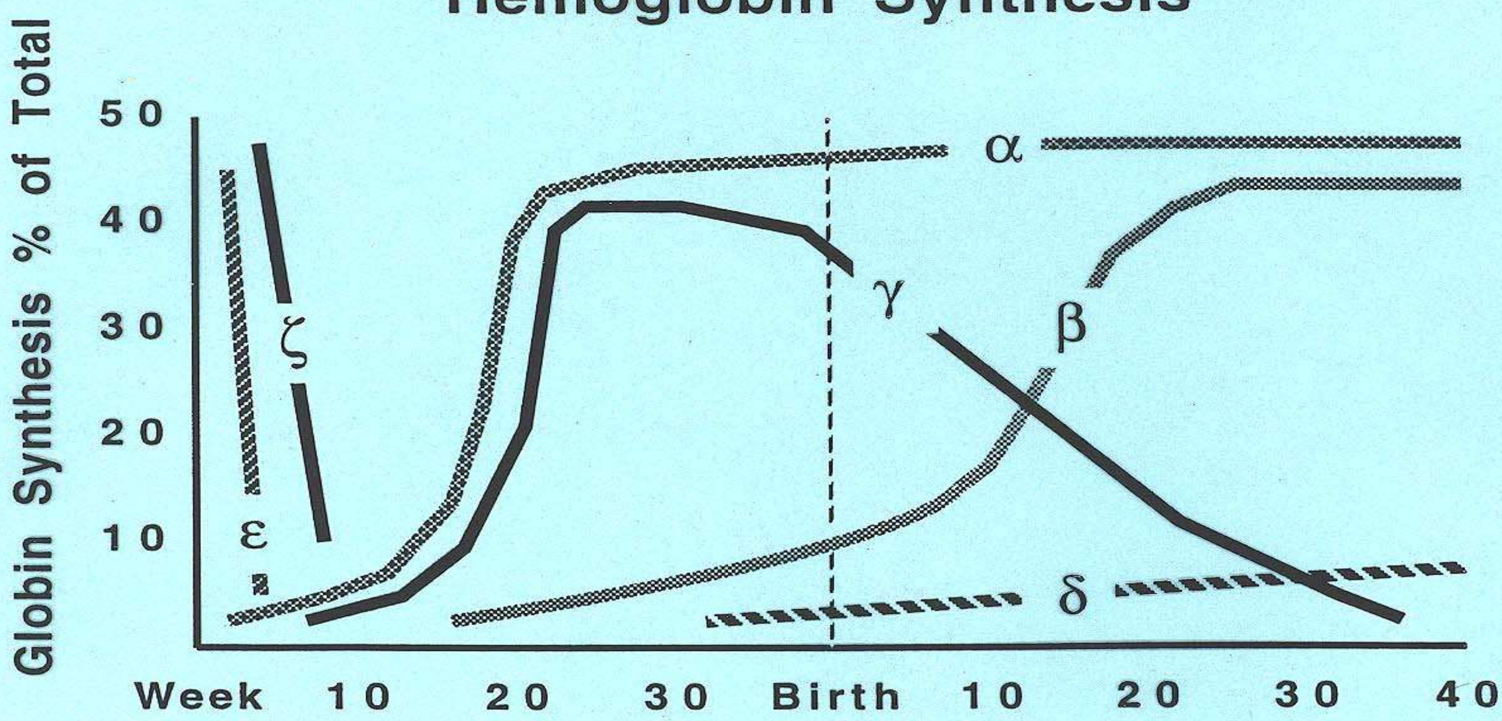
Hemoglobin Types

Hemoglobin Type	Globin Chains
• Hgb A1—97%---------	□ $\alpha2\beta2$
• Hgb A2—2.5%--------	□ $\alpha2\delta2$
• Hgb F — <1%---------	□ $\alpha2\gamma2$
• Hgb H……………………	□ $\beta4$
• Bart's Hgb--------------	□ $\gamma4$
• Hgb S-------------------	□ $\alpha2\beta2^{6\ \mathrm{glu}\rightarrow\mathrm{val}}$
• Hgb C-------------------	□ $\alpha2\beta2^{6\ \mathrm{glu}\rightarrow\mathrm{lys}}$

Genetic basis

- Alpha globins are coded on chromosome 16
  - Two genes on each chromosome
  - Four alleles in each diploid cell
  - **Gene deletions** result in Alpha-Thalassemias
- Also on chromosome 16 are Zeta globin genes—Gower's hemoglobin (embryonic)
- Beta globins are coded on chromosome 11
  - One gene on each chromosome
  - Two alleles in each diploid cell
  - **Point mutations** result in Beta-Thalassemias
- Also on chromosome 11 are Delta (Hgb A2) and Gamma (Hgb F) and Epsilon (Embryonic)

Hemoglobin Synthesis



		Adult	Newborn
$\alpha 2\beta 2$	Hb A	97 %	20 %
$\alpha 2\delta 2$	Hb A2	2.5	<0.5
$\alpha 2\gamma 2$	Hb F	<1	80

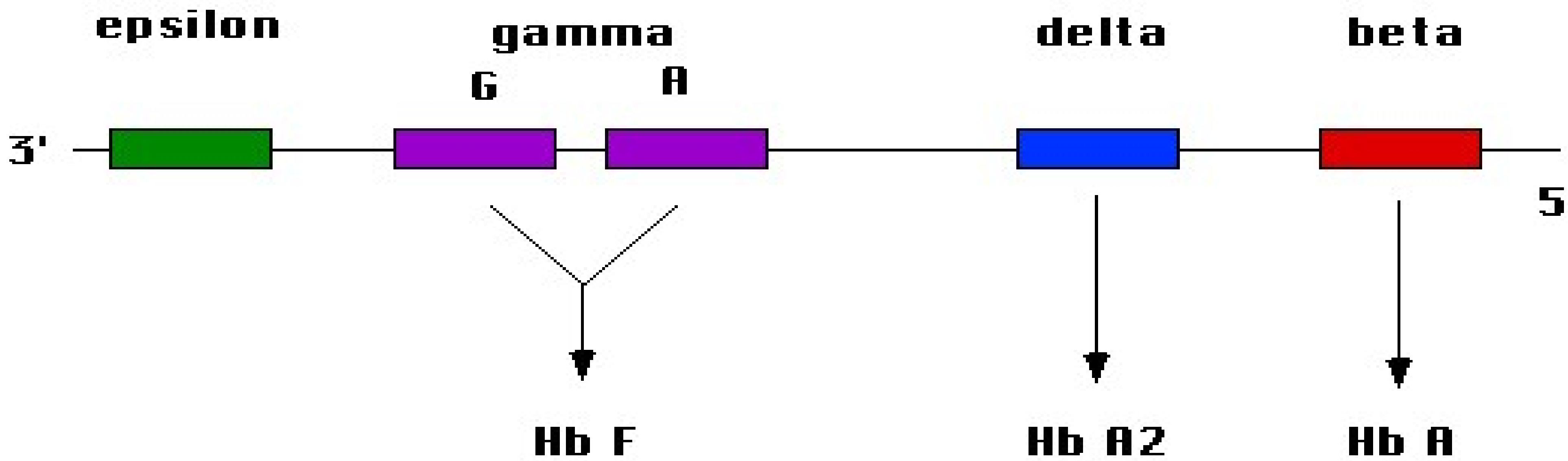
Embryonic:

$\zeta 2\varepsilon 2$ Gower-1

$\alpha 2\varepsilon 2$ Gower-2

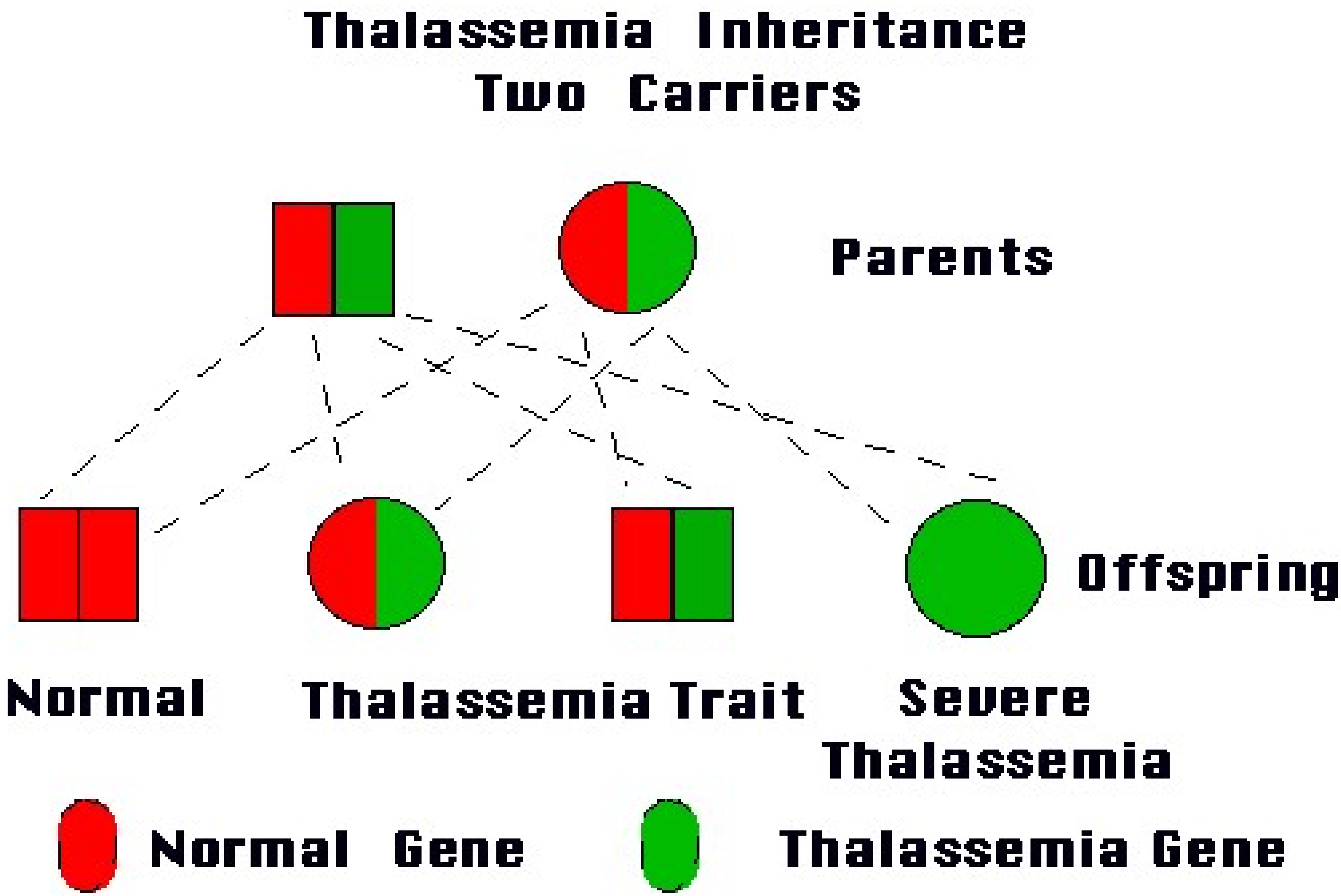
$\zeta 2\gamma 2$ Portland

Beta Globin Gene Cluster
Chromosome 11



Alpha Globin Gene Cluster
Chromosome 16

Thalassemia inheritance



Classification of Alpha Thalassemias

- One deletion—**Silent carrier**; no clinical significance
- Two deletions—α **Thal trait**; mild hypochromic microcytic anemia
- Three deletions—**Hgb H**; variable severity, but less severe than Beta Thal Major
- Four deletions—**Bart's Hgb**; Hydrops Fetalis; In Utero or early neonatal death

- Absence of α chains will result in increase/ excess of **Gamma** globin chains during fetal life and excess β globin chains later in postnatal life.
- Severity of disease depends on number of genes affected.

Alpha Thalassemia genotypes

- Normal $\alpha\alpha/\alpha\alpha$
- Silent carrier $-\alpha/\alpha\alpha$
- Minor(trait) $-\alpha/-\alpha$ (trans)

  $--/\alpha\alpha$ (cis)
- Hb H disease $--/-\alpha$
- Barts hydrops fetalis $--/--$

Beta Thalassemias

- Result from point mutations on genes
- Severity depends on where the hit(s) lie
  - β^0-no β-globin synthesis (total absence)
  - β^+ reduced synthesis (reduced 10-30%)
- Disease results in an overproduction of α-globin chains, which precipitate in the cells and cause splenic sequestration of RBCs
- Erythropoiesis increases, sometimes becomes extramedullary

Classification of Beta Thalassemia

- **Beta thalassemia minor** - heterozygous disorder resulting in mild hypochromic, microcytic hemolytic anemia. **B/ β^{+} or B/β^{0}**
- **Beta thalassemia intermedia** - Severity lies between the minor and major**($\beta^{+/}$ β^{+} or $\beta^{+/}\beta^{0}$)**
- **Beta thalassemia major** - homozygous disorder resulting in severe transfusion-dependent hemolytic anemia**($\beta^{0/}\beta^{0}$)**

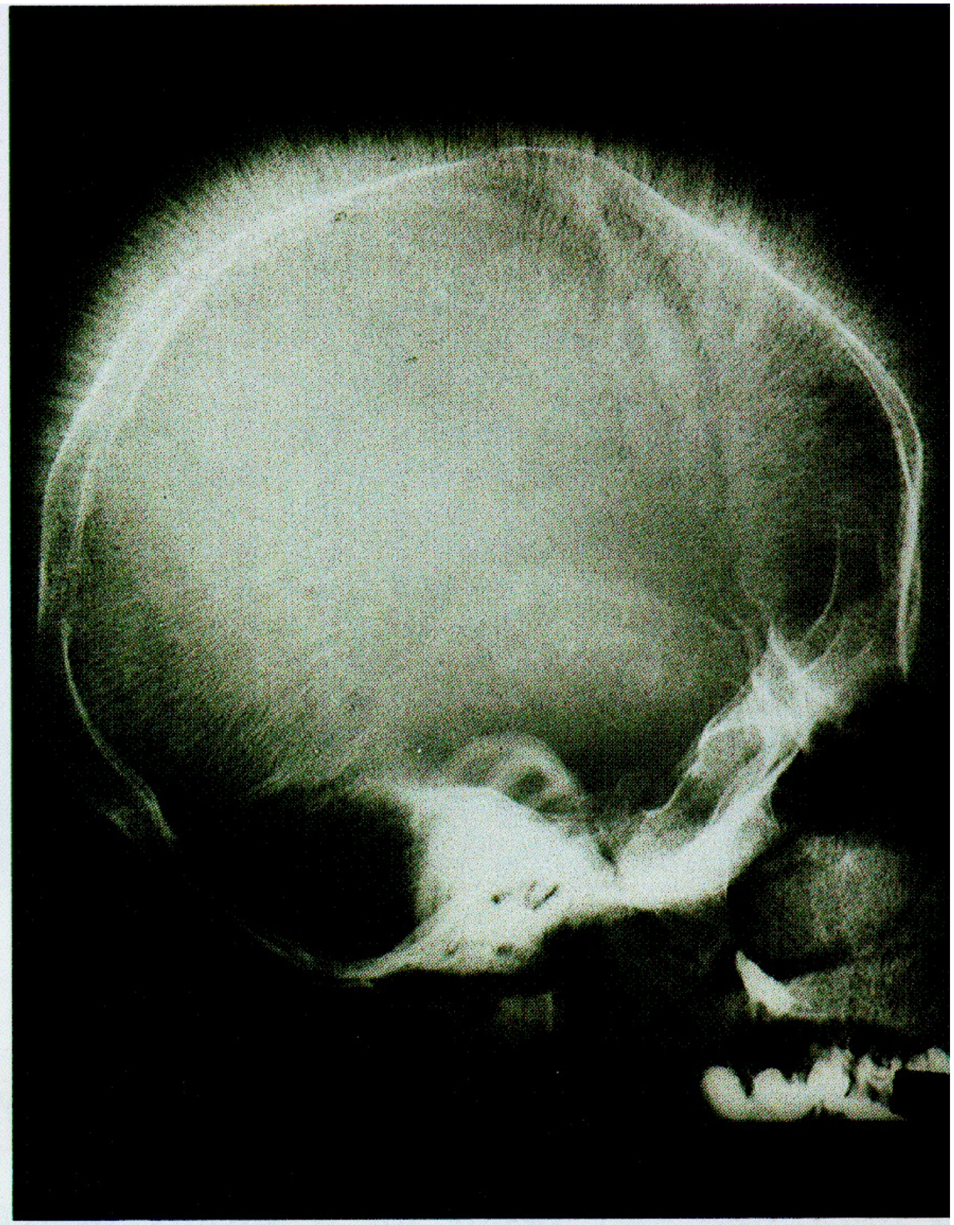
β-Thal--Clinical

- β-Thalassemia Minor
  - Minor point mutation(i.e   regulatory control regions)
  - Minimal anemia; no treatment indicated
- β-Thalassemia Intermedia
  - Homozygous minor point mutation or more severe heterozygote
  - most often do not require chronic transfusions
- β-Thalassemia Major-Cooley’s Anemia
  - Severe gene mutations(i.e nonsense, frame shift)
  - Needs intensive treatment

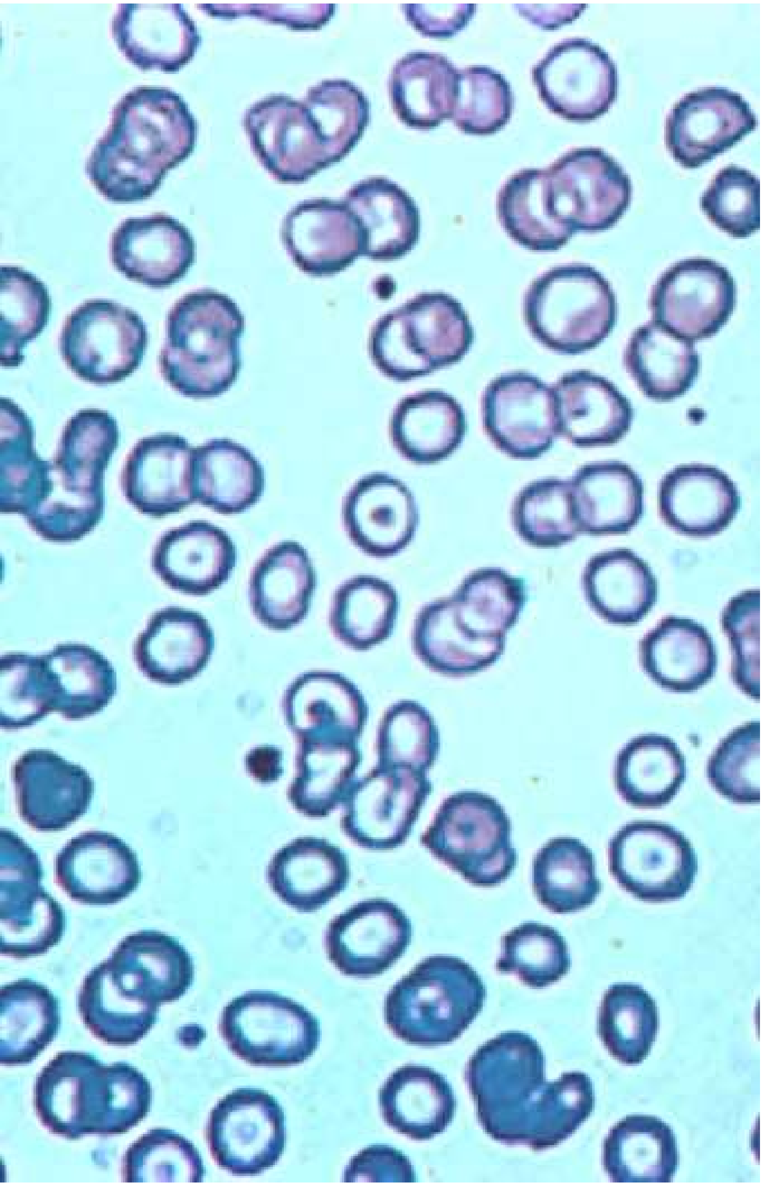
Beta Thalassemia Major

- Reduced or nonexistent production of β-globin
  - Poor oxygen-carrying capacity of RBCs
    - Failure to thrive, poor brain development
  - Increased alpha globin production and precipitation
    - RBC precursors are destroyed within the marrow
- Increased splenic destruction of dysfunctional RBCs
  - Anemia, jaundice, splenomegaly
- Hyperplastic Bone Marrow
  - Ineffective erythropoiesis—RBC precursors destroyed
    - chipmunk face,Poor bone growth, frontal bossing, bone pain
  - Increase in extramedullary erythropoiesis
- Iron overload—increased absorption and transfusions
  - Endocrine disorders, Cardiomyopathy, Liver failure

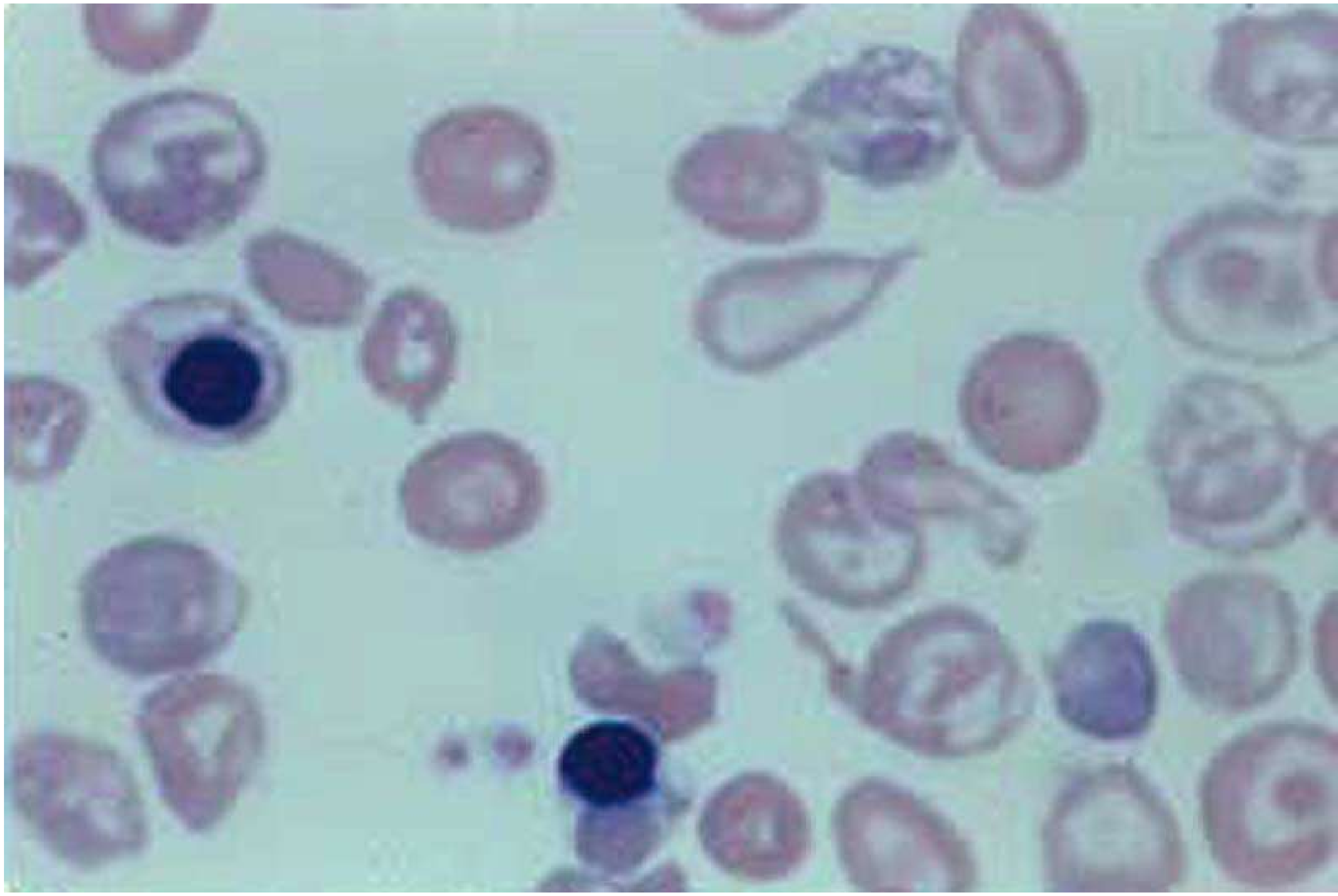
β Thalassemia Major



- Alpha chain accumulation and deposition are more toxic than beta chain accumulation and deposition. Thus beta thalassemia is more severe than alpha thalassemia.



Thalassemia Blood Smears



β-Thalassemia Major—Lab findings

- Hypochromic, microcytic anemia
  - poikiocytosis (Target Cells, nucleated RBCs,teardrop) and anisocytosis
- Reticulocytosis (premature RBCs)
- Hemoglobin electrophoresis shows
  - Increased Hgb A2—delta globin production
  - Increased Hgb F—gamma globin production
- Hyperbilirubinemia

β-Thalassemia Major--Treatment

- Chronic Transfusion Therapy
  - Maximizes growth and development
  - Suppresses the patient’s own ineffective erythropoiesis and excessive dietary iron absorption
  - PRBC transfusions often monthly to maintain Hgb 10-12g/dl
- Chelation Therapy
  - Binds free iron and reduces hemosiderin deposits
  - subcutaneous infusion of deferoxamine, 5 nights/week
  - Start after 1year of chronic transfusions or ferritin>1000 ng/dl
- Splenectomy--indications

β-Thalassemia Major Complications and Emergencies

- Sepsis—Encapsulated organisms
  - *Streptococcus pneumoniae*
- Cardiomyopathy
- Endocrinopathies

On The Horizon

- Oral Chelation Agents (deferiprone and deferosirax)
- Pharmacologically upregulating gamma globin synthesis, increasing Hgb F(hydroxy urea)
  - F hg Carries O2 better than Hgb A2
  - Will help bind α globins and decrease precipitate
- Bone Marrow transplant
- Gene Therapy
  - Inserting healthy β genes into stem cells and transplanting